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18/558,421	11/01/2023	Daniel S. Bates	PA83345US05	4571

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EXAMINER	
HO, WAI-GA DAVID	

ART UNIT	PAPER NUMBER
2872	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/558,421	Applicant(s) Bates et al.	
	Examiner WAI-GA D HO	Art Unit 2872	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 01 November 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-10 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-10 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 11/1/2023 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 1/26/2024.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

2. This office action is in response to the communication filed 11/1/2023.

3. Amendment to the specification, filed 11/1/2023 is acknowledged and accepted.

Information Disclosure Statement

4. The information disclosure statement submitted on 1/26/2025 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Drawings

5. **The drawings are objected to** as failing to comply with 37 CFR 1.84(p)(5) because they include the following reference character not mentioned in the description: α (FIG. 1) . Corrected drawing sheets in compliance with 37 CFR 1.121(d), or amendment to the specification to add the reference character(s) in the description in compliance with 37 CFR 1.121(b) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Specification

6. The disclosure is objected to because of the following informalities:

A. On pg. 1, line 13, "greatly reducing refocusing the eyes" is not proper English.

B. On pg. 1, line 21, "In smaller scale" should read "On a smaller scale"

C. In pg. 1, lines 34-35, "from about 420 nm to about 650 nm the reflective polarizer" needs a comma after "650 nm"

D. On pg. 5, lines 6-7, "Since potentially large amount of information is displayed it…" should read "Since a potentially large of information is displayed, it…"

E. On pg. 9, lines 32-33, "may extends" should read "extends" or "may extend"

F. On pg. 10, line 23, "manufacturers" is missing an apostrophe before the "s"

G. On pg. 10, line 3, "using conventional" should read "using a conventional"

H. On pg. 11, line 8, "400-700 ran" should likely read "400-700 nm"

I. On pg. 11, lines 15-16, it is contradictory and unclear how "a plane of incidence (xz, yz-plane)" is "substantially perpendicular to the first direction (x-axis)". The x-axis lies in the xz-plane.

Appropriate correction is required.

Claim Rejections - 35 USC § 112

7. The following is a quotation of 35 U.S.C. 112(d):

> (d) REFERENCE IN DEPENDENT FORMS.—Subject to subsection (e), a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

The following is a quotation of pre-AIA 35 U.S.C. 112, fourth paragraph:

> Subject to the following paragraph [i.e., the fifth paragraph of pre-AIA 35 U.S.C. 112], a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

8. **Claim 5 is rejected under 35 U.S.C. 112(d)** or pre-AIA 35 U.S.C. 112, 4th paragraph, as being of improper dependent form for failing to further limit the subject matter of the claim upon which it depends, or for failing to include all the limitations of the claim upon which it depends.

9. **Regarding claim 5,** lines 1-4 of this dependent claim recite "**a normalized optical transmittance of the glare trap versus the incident angle comprises a first normalized transmittance peak at a first normalized peak angle with a corresponding normalized full width at half maximum (NFWHM) of less than about 30 degrees".** However, independent claim 1, on lines 20-23, already recited "**an optical transmittance of the glare trap versus the incident angle**... **comprises a first transmittance peak at a first peak angle with a corresponding full width at half maximum (FWHM) of less than about 30 degrees".** Clearly, the limitation of claim 5 mirrors that of the earlier claim 1, except that it is now recited in terms of the **normalized transmittance** and associated values. However, in mathematics, it is generally understood that such normalization involves only a trivial rescaling of the y-axis (**transmittance**), which does not affect any measure taken along the x-axis (i.e. **peak** locations, **FWHM**). Merely reconsidering the **normalized transmittance** and relabeling associated features accordingly is of absolutely no consequence to the system nor to any of its properties. The depending claim 5 thus fails to limit the independent claim 1 on which it depends.

10. Applicant may cancel the claim(s), amend the claim(s) to place the claim(s) in proper dependent form, rewrite the claim(s) in independent form, or present a sufficient showing that the dependent claim(s) complies with the statutory requirements.

Claim Rejections - 35 USC § 103

11. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

12. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

13. **Claims 1, 3, and 5-6 are rejected under 35 U.S.C. 103** as being unpatentable over Otani and Fushimi (US 20170269428 A1, hereinafter "Otani") in view of Jonza et al (US 20020031676 A1, hereinafter "Jonza"), Koike (WO 2021145374 A1), Kato (WO 2020203643 A1), and Smith (US 2851981).

14. **Regarding claim 1,** Otani discloses (see FIGs. 10-11 and ¶s 80-89 regarding a Fourth Exemplary Embodiment – note this incorporates many aspects of a First Exemplary Embodiment described in FIGs. 1-5 and ¶s 22-50) **an optical system** (head-up display 151) **comprising**

A. **a display** (projector 200)**,**

B. **a reflective polarizer** (301)**, and**

C. **a glare trap** (louver optical element 305)**,**

D. **the optical system** (head-up display 151) **configured to display a virtual image** (I) **of an image** (image light B) **emitted by the display** (projector 200) **to a viewer** (driver D) **after the emitted image** (image light B) **is transmitted by the glare trap**

(louver optical element 305) **at least once reflected and at least once transmitted by the reflective polarizer** (301)**,**

E. **the glare trap** (louver optical element 305) **comprising a plurality of spaced apart substantially parallel slats** (light-shielding members 306) **extending along a first direction** (i.e. out of the page in FIGs. 10-11, FIG. 11 annotated below) **and arranged along a different second direction** (horizontally)**, a length L along the first direction and a width W along the second direction,**

F. **wherein the slats** (light-shielding members 306) **in the plurality of spaced apart substantially parallel slats** (light-shielding members 306) **form a plurality of elongated slots therebetween** (see also annotated FIG. 11 below)

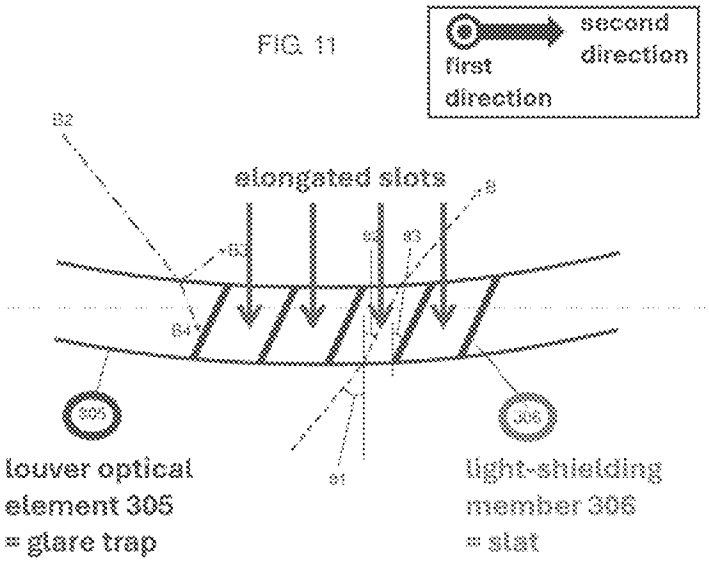


FIG. 11 is annotated to highlight various aspects of Otani's louver optical element 305.

15. Otani does not disclose

A. **L/W ≥ 10,**

B. **elongated slots substantially filled with air,**

C. **for a light incident at an incident angle and a visible wavelength range extending from about 420 nm to about 650 nm:**

i. **for the incident angle of less than about 5 degrees: the reflective polarizer has an average optical reflectance of at least 40% for the incident light having a first polarization state and an average optical transmittance of at least 40% for the incident light having an orthogonal second polarization state, and for each of the first and second polarization states, the glare trap has an average specular optical transmittance of between about 20% to about 80% and an average total optical reflectance of less than about 20%; and**

ii. **for at least one wavelength in the visible wavelength range, an optical transmittance of the glare trap versus the incident angle in a plane of incidence substantially perpendicular to the first direction comprises a first transmittance peak at a first peak angle with a corresponding full width at half maximum (FWHM) of less than about 30 degrees.**

16. Otani and Jonza are commonly related to reflective polarizers for display applications.

17. Jonza discloses general details for fabricating multilayered **reflective polarizer**s (multilayered polymeric sheet 10; see schematics of FIGs. 1(a,b); ¶s 31-40) with tailored properties, i.e. so that the following may be achieved – **for a light incident at an incident angle and a visible wavelength range extending from about 420 nm to about 650 nm:**

A. **for the incident angle of less than about 5 degrees** ("at normal incidence")**: the reflective polarizer** (multilayered polymeric sheet 10) **has an average optical reflectance of at least 40%** (i.e. 90.9%) **for the incident light having a first**

polarization state ("polarized in the stretch direction") **and an average optical transmittance of at least 40%** (i.e. 84.1%) **for the incident light having an orthogonal second polarization state** ("polarized in the non-stretch direction"),

(See Example 8, discussed in ¶s 151-152, and the associated FIG. 28 where:

➔ curve c, for light polarized in the stretch direction, has 9.1% average transmission (corresponding to 90.9% average reflection) across visible wavelengths 400-700nm

➔ curve a, for light polarized in the non-stretch direction, has 84.1% average transmission across the visible wavelengths

Note, per ¶ 130, that absorbance is negligible such that reflection + transmission = 100%)

18. Otani and Koike are commonly related to louvers for glare mitigation in head-up displays

19. Koike discloses (see FIGs. 5, 8 – FIG. 8 annotated/translated below – and ¶s 34-39):

A. **for a light incident at an incident angle and a visible wavelength range:**

i. **for the incident angle of less than about 5 degrees: for each of the first and second polarization states, the glare trap** (louver 70) **has an average total optical reflectance of less than about 20%** (¶s 36 and 39 establish ≥ 90% transmittance of light rays within 10° angle of incidence through louver 70. By conservation of energy, reflectance must be ≤ 10% in this regime. Note also that such scattering properties will generally be **polarization**-independent if no particular (subwavelength, chemical/material) microscopic structure or intentional geometric features are provided to discriminate between polarizations – especially at/near normal incidence, where Fresnel equations lose polarization-dependence)**;**

ii. **for at least one wavelength in the visible wavelength range, an optical transmittance of the glare trap** (louver 70) **versus the incident angle in a plane of incidence** (i.e. the X-Y plane; see ¶ 37) **substantially perpendicular to the first direction** (Z direction) **comprises a first transmittance peak at a first peak angle with a corresponding full width at half maximum (FWHM) of less than about 30 degrees** (as shown in annotated FIG. 8 below)**.**

[図8]

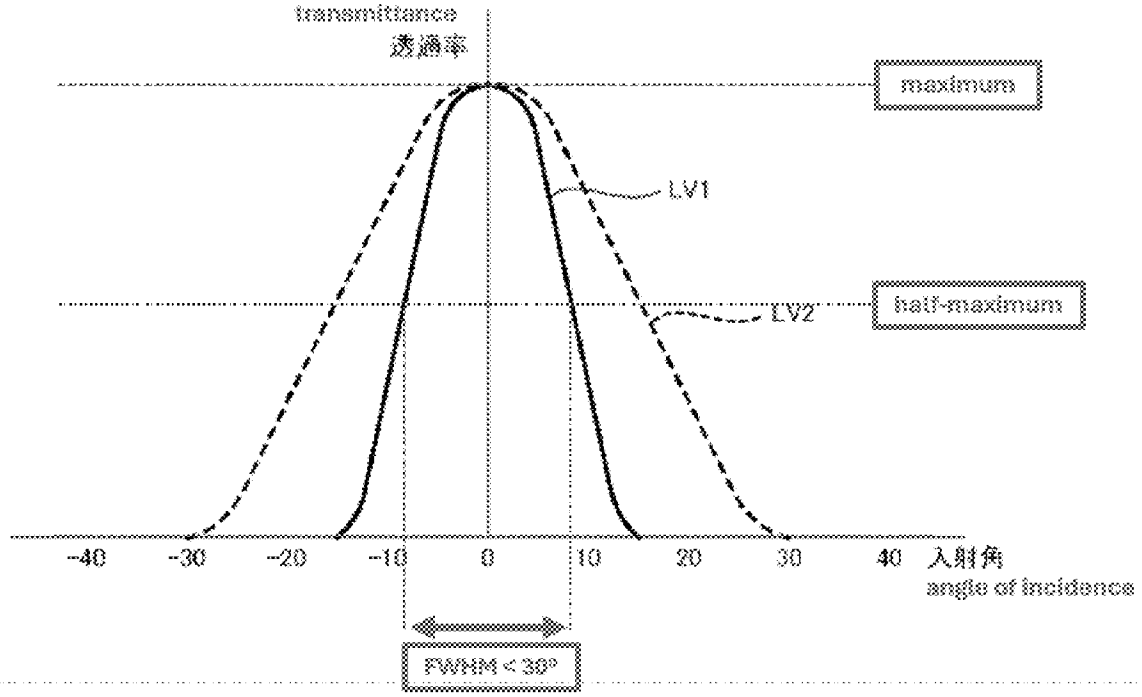


FIG. 8 of Koike is annotated to highlight the FWHM in their transmission plot

20. Otani and Kato are commonly related to louvers for glare mitigation in display devices.

21. Kato discloses (see ¶s 23-34 and FIGs. 1b, 4 regarding the prior art's anisotropic optical film 20 with louver structures/**slats** (columnar structures 23). See also FIGs. 5a and ¶s 35-43, 48-62 regarding Kato's anisotropic optical films 100 whose anisotropic light diffusion layer 110 has louver structures/**slats** (columnar structures 113), analogous to those of the prior art (¶ 39)):

A. **L/W ≥ 10** (¶ 48: "The aspect ratio (= average major axis/average minor axis) [i.e. Kato's ⟨LA⟩/⟨SA⟩ ~ **L/W**] of the multiple columnar structures 113 is 2 or more", encompassing the claimed range)**,**

B. **for a light incident at an incident angle and a visible wavelength range:**

i. **for the incident angle of less than about 5 degrees: for each of the first and second polarization states, the glare trap** (anisotropic light diffusion layer 110) **has an average specular optical transmittance** ("linear transmittance") **of between about 20% to about 80%**

(Note:

➔ per ¶s 59-60, anisotropic light diffusion layers 110/120 (anisotropic optical films 100/150) may have **specular**/linear **transmittance** minima of 10% and maxima of 85%. While such bounds are provided for a broader range of **incident angle**s, note that Kato also provides exemplary transmittance plots for anisotropic optical films 10 and 20 of the prior art in FIG. 3 – showing that one can certainly achieve **specular**/linear **transmittance**s which fall within the claimed range (solid line exceeds **20%** at 0° incidence; i.e. **less than about 5 degrees**)

➔ such scattering properties will generally be **polarization**-independent if no particular (subwavelength, chemical/material) microscopic structure or intentional geometric features are provided to discriminate between polarizations – especially at/near normal incidence, where Fresnel equations lose polarization-dependence

➔ the various teachings that modifying columnar/louver geometries enable control over **specular**/linear transmittance properties as desired (e.g. ¶ 39: "orientation direction... of the columnar structures 113 is... appropriately determined so that the

anisotropic light-diffusing layer 110 has the desired linear transmittance and diffusivity";

¶ 62 on design parameters for tuning linear transmittance properties, etc.)

22. Otani and Smith are commonly related to louver structures.

23. Smith discloses (see FIGs. 1-3; col. 1 line 58 to col. 2 line 17) **elongated slots** (louver openings 13) **substantially filled with air.**

24. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to modify Otani with Jonza in order to design reflective polarizers with desirable and precisely tailored reflection profiles (Jonza ¶s 4, 55).

25. It would have also been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to further modify Otani with Koike in order to achieve desired incident angle-transmittance characteristics and facilitate reduction of stray/undesirable light (Koike ¶s 33-37).

26. It would have then been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to also combine teachings of Otani and Kato in order to tune (linear/specular) transmittance properties for improved reflected light intensities at external angles, and improved visibility and display quality (Kato ¶s 10, 12, 34, 57, 59).

27. It would have lastly been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to combine Otani and Smith in order to produce louver panels with conservative use of materials, lower manufacturing costs, and in a manner conducive to machine operations and mass production (Smith col. 1 lines 34-39, col. 3 lines 5-19).

28. **Regarding claim 3,** modified Otani discloses **the optical system of claim 1.**

29. Smith further discloses (see FIGs. 1-3; col. 1 line 24 to col. 2 line 17) **wherein the glare trap** (louvered panel 10) **comprises a unitary metal sheet comprising a plurality of**

alternating slats (louvers 12) **and elongated through slots** (louver openings 13) **formed in an inner portion thereof leaving a solid perimeter portion** (top, bottom, and side margins 17-20) **substantially surrounding the plurality of alternating slats** (louvers 12) **and elongated through slots** (louver openings 13)**, each elongated through slot** (louver openings 13) **extending between opposing major surfaces of the metal sheet, the slats forming the plurality of spaced apart substantially parallel slats** (louvers 12) **of the glare trap** (louvered panel 10)**.**

30. **Regarding claim 5,** modified Otani discloses **the optical system of claim 1.**

31. Koike further discloses **wherein for the at least one wavelength in the visible wavelength range, a normalized optical transmittance of the glare trap** (louver 70) **versus the incident angle comprises a first normalized transmittance peak at a first normalized peak angle with a corresponding normalized full width at half maximum (NFWHM) of less than about 30 degrees.** (See annotated FIG. 8 above. Examiner notes that the current features parallel those associated with claim 1's **first transmittance peak**, **first peak angle,** and **FWHM** – the only difference being that the current claim 5 is now recited in terms of the **normalized transmittance**. This fails to provide any additional or substantive constraints, however – any transmittance distribution satisfying the referenced aspects of claim 1 will automatically satisfy the current claim. See also ***Claim Rejections - 35 USC § 112*** above.)

32. **Regarding claim 6,** modified Otani discloses (again refer to FIGs. 1-5 and ¶s 22-50, regarding Otani's First Exemplary Embodiment, as well as FIGs. 10-11 and ¶s 80-89, regarding Otani's Fourth Exemplary Embodiment) **a heads-up display (HUD) comprising:**

A. **the optical system of claim 1; and**

B. **a front windshield** (102) **of a vehicle, such that when an image** (image light B) **is emitted by the display** (projector 200)**, the optical system** (head-up display 151) **transmits the emitted image** (image light B) **toward the front windshield** (102)**, the front windshield** (102) **reflects the transmitted image** (image light B) **toward a viewer** (driver D) **in the vehicle, and the viewer** (driver D) **views a virtual image** (I) **of the emitted image** (image light B)**.**

33. **Claim 2 is rejected under 35 U.S.C. 103** as being unpatentable over Otani, Jonza, Koike, Kato, and Smith – as applied to claim 1 above – and in further view of Guo (CN 107288524 A).

34. **Regarding claim 2,** modified Otani discloses **the optical system of claim 1.**

35. Smith further discloses (see FIGs. 1-3; col. 1 line 58 to col. 2 line 17) **wherein the glare trap** (louvered panel 10) **further comprises a frame** (top, bottom, and side margins 17-20)**, the slats** (louvers 12) **in the plurality of spaced apart substantially parallel slats** (louvers 12) **at least partially disposed inside, and at least partially surrounded by the frame** (top, bottom, and side margins 17-20)**, wherein opposite longitudinal ends of each of the slats** (louvers 12) **are secured to the frame** (top, bottom, and side margins 17-20)**.**

36. Modified Otani does not explicitly disclose the **ends of each of the slats are secured to the frame under tension.**

37. Otani and Guo are commonly related to louver structures.

38. Guo discloses (see FIGs. 3, 7; ¶ 31) the **ends of each of the slats** ("louver blades") **are secured to the frame** (guide rail 7) **under tension** (i.e. via telescopic tension spring 9)**.**

39. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to further modify Otani with Guo in order to provide louvers with elastic support with greater tolerance for structural stresses.

40. **Claim 4 is rejected under 35 U.S.C. 103** as being unpatentable over Otani, Jonza, Koike, Kato, and Smith – as applied to claim 1 above – and in further view of Murata and Higashi (US 20070291366 A1, hereinafter "Murata").

41. **Regarding claim 4,** modified Otani discloses **the optical system of claim 1.**

42. Modified Otani does not disclose **wherein for the at least one wavelength in the visible wavelength range, an optical transmittance of the glare trap versus the incident angle in a plane of incidence substantially perpendicular to the second direction comprises first and second transmittance peaks separated by at least 30 degrees, the first and second peaks defining a transmittance valley therebetween having a minimum transmittance less than a greater of the first and second transmittance peaks by at least 1%.**

43. Otani and Murata are commonly related to louvers for glare mitigation in display devices.

44. Murata discloses **wherein for the at least one wavelength in the visible wavelength range, an optical transmittance of the glare trap versus the incident angle in a plane of incidence substantially perpendicular to the second direction comprises first and second transmittance peaks separated by at least 30 degrees, the first and second peaks defining a transmittance valley therebetween having a minimum transmittance less than a greater of the first and second transmittance peaks by at least 1%.** (See ¶s 31-38, 44, 91 and FIGs. 1,3, 5-6 for basic schematic details; light is shone onto anisotropic diffusing medium 1 which is a louver film. The light's incident angles are provided in two orthogonal planes – L and M, or

alternatively/respectively, “short edge” and “long edge” (i.e. with respect to edges of the sample), corresponding to **first** and **second directions** – and **transmittance**s for each are measured as functions of incident angle.

These transmittance curves are provided in FIGs. 13-17 for several specific examples – each depicting **transmittance valley**s with minima as much as tens of percent below larger **transmittance peak**s that are separated by several tens of angles of incidence (see also annotated FIG. 13 below).

Examiner thus finds that, when taken in combination with the above-cited works (note that Murata belongs to a larger body of work that also includes above-cited Kato), there are sufficiently enabling details to produce a louver structure whose transmittance spectra contains the claimed **peak/valley** features.)

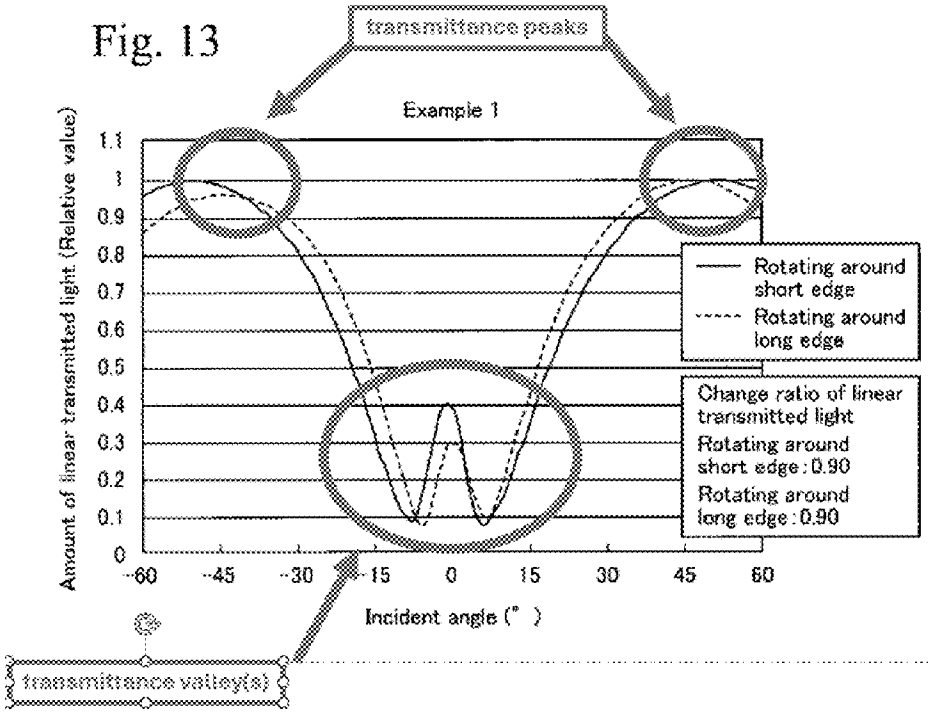


FIG. 13 of Murata is annotated to highlight transmittance peaks and valleys

45. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to further combine Otani with Murata in order to improve anisotropic diffusion characteristics for display applications (Murata ¶ 8-10).

46. **Claim 7 is rejected under 35 U.S.C. 103** as being unpatentable over Otani and Fushimi (US 20170269428 A1, hereinafter "Otani") in view of Koike (WO 2021145374 A1), Kato (WO 2020203643 A1), and Smith (US 2851981).

47. **Regarding claim 7,** Otani discloses **a light control film** (louver optical element 305) **comprising** (refer again to annotated FIG. 11 above)

A. **a plurality of spaced apart substantially parallel light blocking regions** (light-shielding members 306) **extending along a first direction** (i.e. out of the page in FIGs. 10-11, FIG. 11 annotated above) **and arranged along an orthogonal second direction** (horizontally)**, each light blocking region** (light-shielding members 306) **comprising a slat extending continuously along the length of the light blocking region, the slat having a length L along the first direction and a width W along a width direction of the slat,**

B. **wherein the light blocking regions** (light-shielding members 306) **form a plurality of elongated through slots therebetween.**

48. Otani does not disclose:

A. that the slats are **metal slats**

B. **L/W > 10,**

C. **slots substantially filled by air**

D. **wherein for light incident on the light control film and for at least one wavelength in a visible wavelength range extending from about 420 nm to about 650 nm, an optical transmittance of the light control film versus an incident angle of the incident light in a plane of incidence substantially perpendicular to the first direction comprises a first transmittance peak at a first peak angle with a corresponding full width at half maximum (FWHM) of less than about 25 degrees.**

49. Otani and Koike are commonly related to louvers for glare mitigation in head-up displays

50. Koike discloses (see FIGs. 5, 8 – FIG. 8 annotated/translated above – and ¶s 34-39) **wherein for light incident on the light control film and for at least one wavelength in a visible wavelength range extending from about 420 nm to about 650 nm, an optical transmittance of the light control film** (louver 70) **versus an incident angle of the incident light in a plane of incidence** (i.e. the X-Y plane; see ¶ 37) **substantially perpendicular to the first direction** (Z direction) **comprises a first transmittance peak at a first peak angle with a corresponding full width at half maximum (FWHM) of less than about 25 degrees** (as shown in annotated FIG. 8 above)**.**

51. Otani and Kato are commonly related to louvers for glare mitigation in display devices.

52. Kato discloses **L/W > 10** (¶ 48: "The aspect ratio (= average major axis/average minor axis) [i.e. Kato's ⟨LA⟩/⟨SA⟩ ~ **L/W**] of the multiple columnar structures 113 is 2 or more", encompassing the claimed range)

53. Otani and Smith are commonly related to louver structures.

54. Smith discloses (see FIGs. 1-3; col. 1 line 58 to col. 2 line 17):

A. that the slats are **metal slats**

B. **slots substantially filled by air**

55. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to modify Otani with Koike in order to achieve desired incident angle-transmittance characteristics and facilitate reduction of stray/undesirable light (Koike ¶s 33-37).

56. It would have also been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to further combine teachings of Otani and Kato in order to tune (linear/specular) transmittance properties for improved reflected light intensities at external angles, and improved visibility and display quality (Kato ¶s 10, 12, 34, 57, 59).

57. It would have then been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to also combine Otani and Smith in order to produce louver panels with conservative use of materials, lower manufacturing costs, and in a manner conducive to machine operations and mass production (Smith col. 1 lines 34-39, col. 3 lines 5-19)

58. **Claims 8 and 10 are rejected under 35 U.S.C. 103** as being unpatentable over Otani and Fushimi (US 20170269428 A1, hereinafter "Otani") in view of Smith (US 2851981), First International Computer Inc. (NPL entitled *AR HUD Design Solution*, hereinafter "FIC"), and Kato (WO 2020203643 A1).

59. **Regarding claim 8,** Otani discloses (see FIGs. 10-11 and ¶s 80-89 regarding a Fourth Exemplary Embodiment – note this incorporates many aspects of a First Exemplary Embodiment described in FIGs. 1-5 and ¶s 22-50) **a heads-up display (HUD)** (151) **comprising:**

A. **a display** (projector 200)**,**

B. **a reflective polarizer** (301)**, and**

C. **a glare trap** (louver optical element 305)**,**

D. **the HUD** (151) **configured to display a virtual image** (I) **of an image** (image light B) **emitted by the display** (projector 200) **to a viewer** (driver D) **after the emitted image** (image light B) **is transmitted by the glare trap** (louver optical element 305) **and at least once reflected and at least once transmitted by the reflective polarizer** (301),

E. **the HUD comprising an optic axis extending between the display and the viewer, the glare trap** (louver optical element 305) **configured to be positioned so that the emitted image** (image light B) **propagates a distance D along the optic axis between the glare trap** (louver optical element 305) **and the viewer** (driver D) (see annotated FIG. 10 below. Examiner will note that the distance between a driver's head and the front windshield, plus any additional distances associated with the items in the front of a car, will generally be of $\mathcal{O}[1\text{ m}]$ in most passenger vehicles. Examiner will thus take $\boldsymbol{D} = 1\text{ m}$ as a reasonably accurate estimate for the usual configuration),

FIG. 10

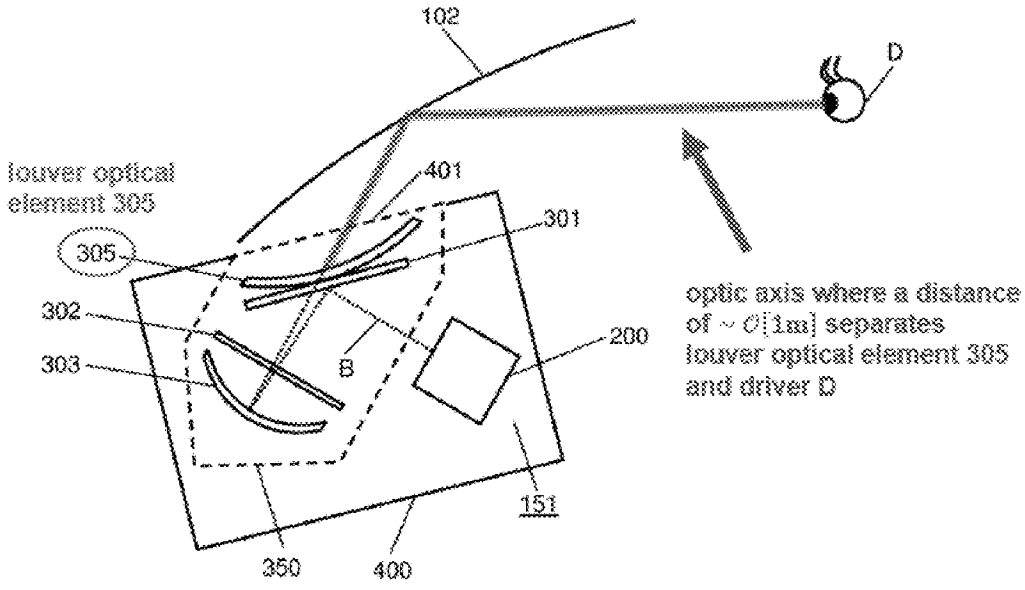


FIG. 10 of Otani is annotated to highlight the optic axis

F. **the glare trap** (louver optical element 305) **comprising a plurality of spaced apart substantially parallel slats** (light-shielding members 306) **extending along a first direction** (i.e. out of the page in FIGs. 10-11, FIG. 11 annotated above) **and arranged along a different second direction** (horizontally)**, each slat comprising a width W along the second direction and a thickness H along a third direction orthogonal to the first and second directions.**

60. Otani does not specifically disclose**:**

A. that the slats are **metal slats**

B. **wherein H in units of mm is not greater than:**

$$n\frac{3\times 10^{-4}\cdot D - W}{\phi}$$

and is not less than:

$$n\frac{W}{0.35}$$

where n is an index of refraction of a material substantially filling the spaces between the slats, ϕ is a half of a field angle of an edge of the virtual image in units of radians, and D and W are in mm.

61. Otani and Smith are commonly related to louver structures.

62. Smith discloses (see FIGs. 1-3; col. 1 line 58 to col. 2 line 17) that the slats are **metal slats.**

63. Otani and FIC are commonly related to head-up displays for automotives.

64. FIC discloses that **ϕ is a half of a field angle of an edge of the virtual image in units of radians** (see pg. 2: "For example, 5°[= 0.087…] FOV Portion… 20°[= 0.349…] FOV Portion", corresponding to a range for **half of** the **field angle**: $0.044 \lesssim \phi \lesssim 0.175$ **).**

65. Otani and Kato are commonly related to louvers for glare mitigation in display devices.

66. Kato, in combination with the above-cited art, discloses **wherein H in units of mm is not greater than:**

$$n\frac{3\times 10^{-4}\cdot D - W}{\phi}$$

and is not less than:

$$n\frac{W}{0.35}$$

where n is an index of refraction of a material substantially filling the spaces between the slats, and D and W are in mm. (Note the following:

➔ In ¶ 98, Kato establishes refractive indices of $n \simeq 1.5$ for their materials for their anisotropic optical films)

➔ In ¶ 51, Kato establishes a preferred range of $0.5\ \mu m \leq W \leq 5\ \mu m (= 5\times 10^{-6} m)$ for the minor axis or **width** of columnar structures 113 (i.e. **slats)**

➔ The above two points thus provide a lower bound on the thickness H

$$n\frac{W}{3.5} \simeq 1.5 * \frac{0.5\ \mu m}{3.5} \simeq 2.14\ \mu m$$

and – together with the $D = 1$ m rationalized with respect to Otani above, and the lower bound on $\phi = 0.044$ cited from FIC above – an upper limit of

$$n\frac{3\times 10^{-4}\cdot D - W}{\phi} \simeq 1.5 * \frac{3\times 10^{-4}\cdot (1m) - (5\times 10^{-6} m)}{0.044} \simeq 0.01\ m = 10\ mm$$

when applying the claimed formulas.

➔ In ¶ 57, Kato establishes a preferred range of $10\ \mu m \lesssim H \lesssim 200\ \mu m$ for the **thickness** of the columnar structures 113, which falls squarely within the range of $2.14\ \mu m \lesssim H \lesssim 10 mm$ derived separately above using the claimed formulas.)

67. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to combine Otani and Smith in order to produce louver panels with conservative use of materials, lower manufacturing costs, and in a manner conducive to machine operations and mass production (Smith col. 1 lines 34-39, col. 3 lines 5-19).

68. It would have also been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to combine teachings of Otani and FIC in order to provide bright images that are readable under natural sunlight and provide depth perception (FIC pg. 1)

69. It would have then been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to also combine teachings of Otani and Kato in order to tune (linear/specular) transmittance properties for improved reflected light intensities at external angles, and improved visibility and display quality (Kato ¶s 10, 12, 34, 57, 59).

70. **Regarding claim 10,** modified Otani discloses **the HUD of claim 8.**

71. FIC further discloses **wherein ϕ is less than about 0.35 radians.** (see pg. 2: "For example, 5°[= 0.087…] FOV Portion… 20°[= 0.349…] FOV Portion", corresponding to a range: $0.044 \lesssim \phi \lesssim 0.175$)

72. **Claim 9 is rejected under 35 U.S.C. 103** as being unpatentable over Otani in view of Smith, FIC, and Kato – as applied to claim 8 above – and in further view of Weber et al (US 20040135742 A1, hereinafter "Weber")

73. **Regarding claim 9,** modified Otani discloses **the HUD of claim 8.**

74. Modified Otani does not disclose the **further comprising a second reflective polarizer configured to receive the transmitted image and reflect a portion of the received image toward the eye.**

75. Otani and Weber are commonly related to reflective polarizers in head-up displays

76. Weber discloses (see FIG. 2) the **further comprising a second reflective polarizer** (20) **configured to receive the transmitted image** (light 14) **and reflect a portion** (light/portion 14a) **of the received image** (light 14) **toward the eye** (viewer 22)**.**

77. It would have therefore been obvious for one of ordinary skill in the art, before the effective filing date of the claimed invention, to combine Otani with Weber in order to avoid image distortion resulting from reflections at the outer surface of the windshield and provide compatibility with polarized sunglasses (¶s 7-9, 37).

Conclusion

78. Any inquiry concerning this communication or earlier communications from the examiner should be directed to WAI-GA D. HO whose telephone number is (571)270-1624. The examiner can normally be reached Monday through Friday, 10AM - 6PM E.T..

79. Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

80. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Stephone Allen can be reached at (571) 272-2434. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

81. Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

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